Find the intervals on which each function is continuous.

5) $f(x) = \frac{x^2}{2x + 4}$

6) $f(x) = \begin{cases} -\frac{x}{2} - \frac{7}{2}, & x \le 0 \\ -x^2 + 2x - 2, & x > 0 \end{cases}$

7) $f(x) = -\frac{x^2 - x - 12}{x + 3}$

8) $f(x) = \frac{x^2 - x - 6}{x + 2}$

Determine if each function is continuous. If the function is not continuous, find the *x*-axis location of and classify each discontinuity.

9) $f(x) = -\frac{x^2}{2x + 4}$

10) $f(x) = \frac{x + 1}{x^2 - x - 2}$

11) $f(x) = \frac{x + 1}{x^2 + x + 1}$

12) $f(x) = -\frac{x^2}{x - 1}$

13) $f(x) = \begin{cases} x^2 - 4x + 3, & x \ne 0 \\ 3, & x = 0 \end{cases}$

14) $f(x) = \begin{cases} -x^2, & x \ne 1 \\ 0, & x = 1 \end{cases}$

Critical thinking questions:

15) Give an example of a function with discontinuities at $x = 1$, 2, and 3.

16) Of the six basic trigonometric functions, which are continuous over all real numbers? Which are not? What types of discontinuities are there?